

FortiGate Firewall Configuration Report

Network Segmentation | Security Policies | Web & App Filtering

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Device	FortiGate Next-Generation Firewall (NGFW)
FortiOS Version	Enterprise release (version withheld)
Scope	Network Segmentation, Firewall Policies, Web Filter, Application Control
Report Type	Technical Reference
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1. Introduction

This document is a technical proof-of-work and reference for a FortiGate Next-Generation Firewall (NGFW) configuration. The configuration was performed in a live enterprise environment running FortiOS, deployed at a corporate office location. All organisation-identifying details have been redacted from this public version.

The objective was to establish a secure, team-segmented network providing each department with appropriately controlled internet access. The project spanned static IP assignment, team-based address grouping, web filtering, application control, and consolidated firewall policy creation.

2. Environment Overview

Firewall Device	FortiGate Next-Generation Firewall (NGFW)
FortiOS Version	Enterprise release (withheld)
Hostname	Withheld
Management URL	https://[firewall_ip] — internal only
LAN Subnet	192.168.x.x/24 (details withheld)
WAN	Virtual WAN Link (SD-WAN)
DNS	Cloudflare (1.1.1.1 / 1.0.0.1), Google (8.8.8.8 / 8.8.4.4)
Teams Configured	7 department groups

3. Phase 1 — Static IP Assignment & Network Segmentation

3.1 Approach

All team devices were migrated from DHCP to static IPs. This is a prerequisite for reliable firewall rules — dynamic IPs change on reconnection, making policy enforcement unpredictable. Each device was verified for connectivity post-assignment.

3.2 IP Address Allocations by Team

The following table shows the address object structure. Specific IP values are withheld for security:

Team / Group	IP Range	Address Object Name	Type
Team A	192.x.x.x – 192.x.x.x	TEAM_A	IP Range
Team B	192.x.x.x – 192.x.x.x	TEAM_B	IP Range
Team C	192.x.x.x – 192.x.x.x	TEAM_C	IP Range
Team D	192.x.x.x – 192.x.x.x	TEAM_D	IP Range
Team E	192.x.x.x – 192.x.x.x	TEAM_E	IP Range
Team F	192.x.x.x – 192.x.x.x	TEAM_F	IP Range
Team G	192.x.x.x – 192.x.x.x	TEAM_G	IP Range

3.3 Screenshot — FortiGate Policy & Objects > Addresses

The screenshot below (redacted) confirms the address objects configured for each team group:

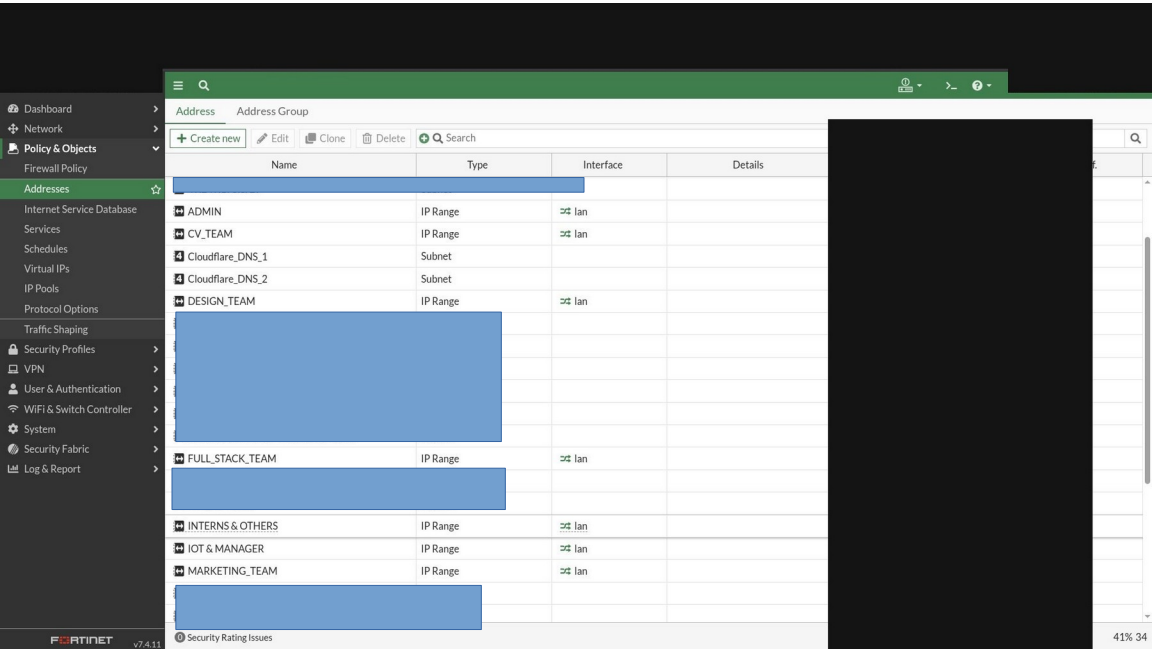


Figure 1: FortiGate Policy & Objects > Addresses — Team IP ranges as address objects [sensitive data redacted]

4. Phase 2 — Web Filter Profiles

4.1 Overview

Separate Web Filter profiles were created per team under Security Profiles > Web Filter, configuring URL category blocking, custom allow/block lists, and FortiGuard enforcement per team's requirements.

Profile Name	Purpose / Notes
team_a_webfilter	URL category controls for Team A
team_b_webfilter	URL category controls for Team B
team_c_webfilter	Web access profile for Team C
general_block	Baseline blocking for unmatched traffic
team_d_webfilter	Access restrictions for Team D
team_e_webfilter	Web access profile for Team E

4.2 Screenshot — Web Filter Profiles

The screenshot below shows the full list of web filter profiles created:

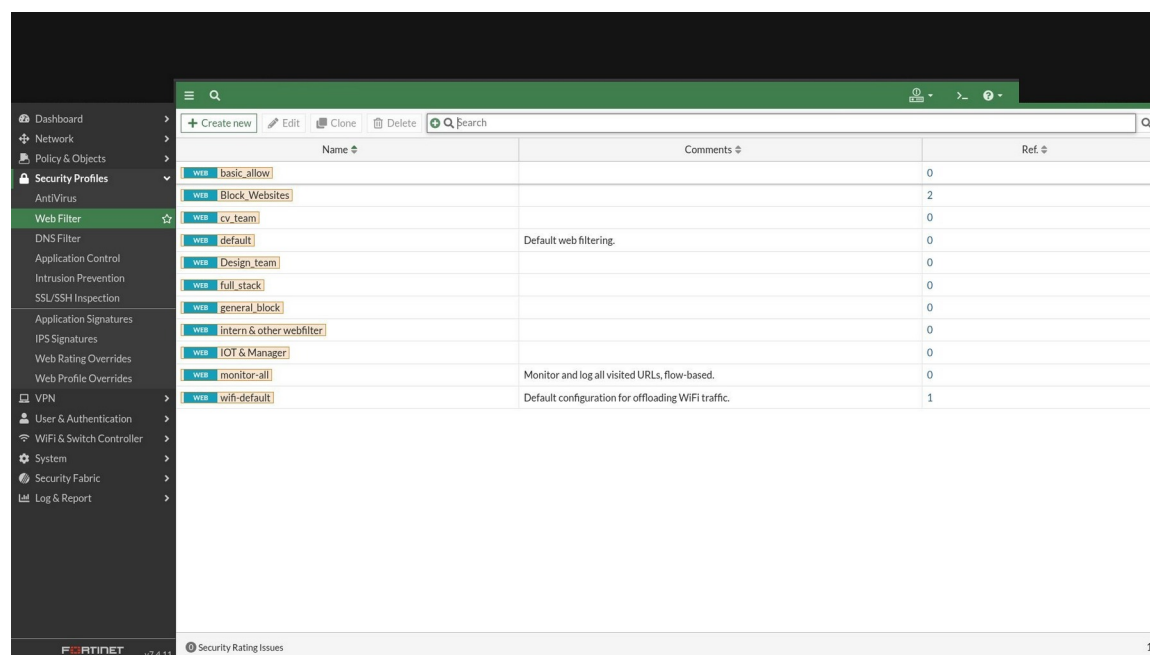


Figure 2: FortiGate Security Profiles > Web Filter — Team-specific profiles [sensitive data redacted]

4.3 Configuration Highlights

- FortiGuard URL category enforcement per team — social networking, gaming, adult content blocked where appropriate
- Custom URL allow/block lists added for team-specific exceptions
- general_block profile applied as fallback for unclassified traffic

5. Phase 3 — Application Control Profiles

5.1 Overview

Application Control profiles provide deep-packet inspection at the application layer, identifying specific applications by traffic signatures regardless of port or protocol.

Profile Name	Comment / Purpose
team_a_app_control	Blocks social media, gaming, and non-productive apps
team_b_app_control	Productivity-focused application ruleset
team_c_app_control	Blocks unauthorised communication tools
team_d_app_control	Application control for Team D
team_e_app_control	Restricts streaming and entertainment apps
team_f_app_control	Limited social media access where business-required

5.2 Screenshot — Application Control Profiles

The screenshot below confirms the application control profiles created per team:

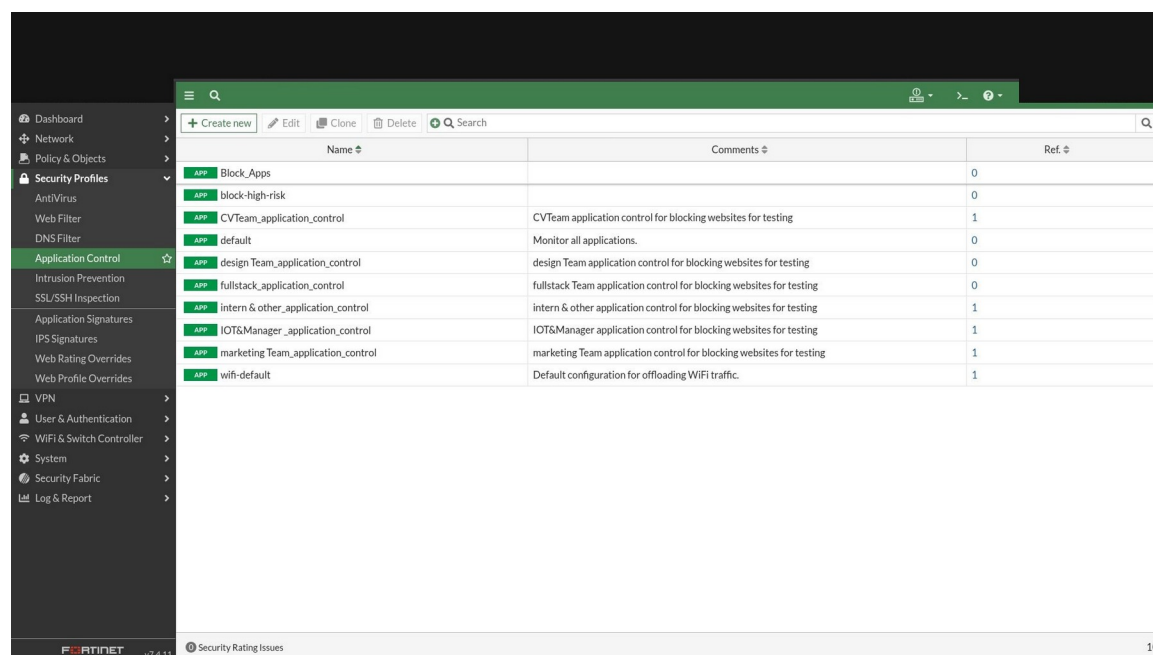


Figure 3: FortiGate Security Profiles > Application Control — Team-specific profiles [sensitive data redacted]

5.3 Configuration Highlights

- Application signatures used to block social media, gaming, and unauthorised communication tools per team
- DPI via SSL certificate-inspection to inspect HTTPS application traffic

6. Phase 4 — Firewall Policies

6.1 Policy Architecture

Individual firewall policies were created per team, binding each source address group (team IP range) to the internet via Virtual WAN Link with corresponding security profiles.

All team policies: ACTION: ACCEPT | NAT: Enabled | SSL Inspection: certificate-inspection | Schedule: always | Service: ALL

6.2 Policy Summary

Policy Name	Source Address	Web Filter	App Control
Admin_Policy	ADMIN	—	—
Team_A_Policy	TEAM_A	team_a_webfilter	team_a_app_control
Team_B_Policy	TEAM_B	team_b_webfilter	team_b_app_control
Team_C_Policy	TEAM_C	team_c_webfilter	team_c_app_control
Team_D_Policy	TEAM_D	team_d_webfilter	team_d_app_control
Team_E_Policy	TEAM_E	team_e_webfilter	team_e_app_control
Team_F_Policy	TEAM_F	—	team_f_app_control
Intern_Policy	INTERNS	intern_webfilter	intern_app_control

6.3 Screenshot — Firewall Policy

The screenshot below shows the complete policy table with security profiles applied per team:

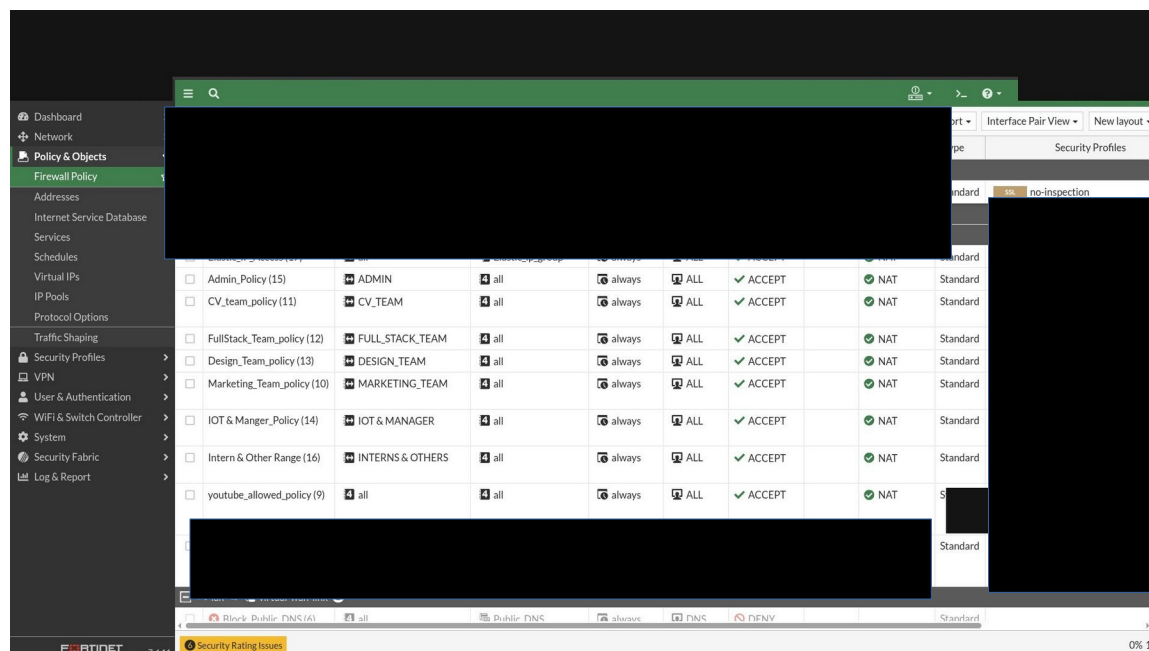


Figure 4: FortiGate Policy & Objects > Firewall Policy — All team policies [sensitive data redacted]

6.4 Key Policy Notes

- ALLOW_FGT_DNS — Permits DNS to Virtual WAN; no SSL inspection (DNS cannot be decrypted)
- Block_Public_DNS — DENY policy blocking public DNS bypass (e.g. 8.8.8.8), forcing DNS through FortiGate filter
- YouTube Policy — Allows YouTube with full security scanning (Antivirus + IPS + SSL inspection)
- LAN_to_Internet — Catch-all: unmatched traffic still subject to baseline web filter

7. Configuration Workflow Summary

End-to-end steps completed:

①	DHCP to Static IP Migration — Configured static IPs per team range; verified connectivity
②	Address Object Creation — IP Range objects per team in Policy & Objects > Addresses
③	Web Filter Profile Creation — Per-team profiles with URL category blocking and FortiGuard enforcement
④	Application Control Profile Creation — Per-team profiles with signature-based app blocking
⑤	Firewall Policy Creation & Testing — Team policies with NAT, SSL inspection; tested from representative devices
⑥	Verification — Confirmed block page for blocked URLs; verified Forward Traffic logs matched per source IP

8. Outcome & Results

Network Segmentation	Seven team groups assigned dedicated static IP ranges in a logically segmented LAN
Tailored Access Control	Each team governed by a dedicated firewall policy with team-specific profiles
HTTPS Inspection	SSL certificate-inspection across all team policies for encrypted traffic visibility
DNS Security	Public DNS bypass blocked, forcing DNS through FortiGate's DNS filter
Baseline Protection	Catch-all policy ensures unmatched traffic is still filtered

